MIL-R-43658A
19 September 1983
SUPERSEDING
MIL-R-43658
25 July 1969

MILITARY SPECIFICATION

RACK SET, BAKING FOR CABINETS, GASOLINE, FIELD

RANGE M1937 AND M1959

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a baking rack set for use with the Cabinets, Gasoline, Field Range M1937 and M1959.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

* SPECIFICATION

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MS 35338 - Washer, Lock-Spring, Helical, Regular (Medium) Series

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7360

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DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

5-11-353 - Rack Set, Baking, for Cabinets, Gasoline, Field Range M1937 and M1959

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

THE ALUMINUM ASSOCIATION

Designation System for Aluminum Finishes 3rd Edition-1966

(Application for copies should be addressed to the Aluminum Association, 420 Lexington Ave., New York, NY 10017.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order or precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.2 Materials (see 6.3) and components. Materials and components shall be as specified on Drawing 5-11-353.

3.3 Design and construction. The baking rack set shall consist of a top, middle and bottom rack, and shall be constructed as shown on Drawing 5-11-353.

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3.4 Finish. The finish of the aluminum on the racks shall conform to designation M 10 specified in Aluminum Association "Designation System for Aluminum Finishes", 3rd Edition.

3.5 Marking for identification. Each rack shall be stamped with the letters U.S. and the manufacturer's name, trade name, or trademark in the size and location shown on Drawing 5-11-353. Adjacent surfaces and surface opposite the markings shall show no signs of penetration, unevenness or distortion.

3.6 Workmanship. The finished racks shall be free of cracks, dents, malformation or deep scratches. All rough and sharp edges, burrs, slivers, grease, or other foreign matter shall be removed.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.1, 4.4.2, and 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

* 4.4.2 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of baking rack sets. The sample unit shall be one baking rack set. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects.

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TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Construction and workmanship	Any component fractured, split, punctured, dented, bowed, or malformed	X	
	Any burrs, slivers, or rough or sharp edges	X	
	Any component missing	X	
	Scale, grease or foreign matter		X
Marking for identification	Missing, incorrect, or illegible		X
	Not size specified		X
	Signs of penetration, unevenness, or distortion on adjacent or opposite surfaces		X

* 4.4.3 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified on the drawing. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of baking rack sets. The sample unit shall be one baking rack set. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

* 4.4.4 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in term of defects per hundred units, shall be 2.5.

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Contents	Number of cases per container is more or less than specified

* 4.4.5 Palletization examination. An examination shall be made to determine that palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred unit, shall be 6.5.

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Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limit
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each set of racks shall be unit packed in a snug-fitting fiberboard box conforming to style FPF, type CF, class SW, or type SF, class domestic of PPP-B-636. The racks shall be placed in the box with the middle rack positioned between the top and bottom rack. Fiberboard pads, made of the same material as the box with dimensions approximately 1/8 inch less than the length and width of the box, shall be placed between the racks to prevent contact. Box closure shall be in accordance with the appendix of PPP-B-636.

* 5.1.2 Commercial. Each rack set shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Four sets of racks of one type only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636, except that the weight limitations shall be waived. Each shipping container shall be closed and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

5.2.2 Level B packing. Four sets of racks of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636, except that the weight limitations shall be waived. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636.

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5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2) the shipping containers shall be a grade V3c, V3s, or V4s fiberboard box fabricated and closed in accordance with PPP-B-636.

* 5.2.3 Commercial packing. Rack sets preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), rack sets packed as specified in 5.2 shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract, unit packs, and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The rack set, when placed in the cabinets, will provide shelves for baking bread and pastry products. The rack set is used with the Cabinets, Gasoline, Field Range M1937 and M1959.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When first article is required (see 3.1, 4.3, and 6.4).
c. Selection of the applicable levels of packaging and preservation (see 5.1 and 5.2).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
e. When palletization is required (see 5.3).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one complete rack set. The contracting officer should include specific instructions in all acquisition documents, regarding arrangement for inspection and approval of the first article.

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6.5 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Review activities:

Army - MD
Navy - YD
Air Force - 84

User activity:

Navy - MC

Preparing activity:

Army - GL

Project No. 7360-0147

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-43658A	Rack Set, Baking For Cabinets, Gasoline, Field Range M1937 And M1959

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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